

EEPROM Interface for Beginners

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Electronic systems today incorporate a number of peripheral ICs that have to communicate with each other and the outside world. To maximise hardware efficiency and simplify circuit design, Philips developed a simple bidirectional two-wire communication interface between components residing on the same circuit board. This interface is referred to as two-wire interface (TWI) or FC (inter-integrated circuit).

Presented here is a circuit for interfacing AT24C02 EEPROM with P89V51RD2 microcontroller using FC protocol. AT24C02 EEPROM is a non-

volatile memory often used for interfacing with microcontrollers. That is, data is not lost even when power fails. A microcontroller can read the last saved data from the EEPROM when power resumes.

Circuit and working

Fig. 1 shows the circuit of EEPROM interface for beginners. It comprises a regulator 7805 (IC1), microcontroller P89V51RD2 (IC2), EEPROM AT24C02 (IC3), an LCD module and a few discrete components. The 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 9 V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC1. Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and resistor R1 limits the current through LED1.



Microcontroller. The P89V51RD2 is an eight-bit 80C51 microcontroller with 64kB Flash and 1024 bytes of data RAM. It has three 16-bit timers/counters, programmable watchdog timer, eight interrupt sources with four priority levels, serial peripheral interface (SPI) and an enhanced UART, programmable counter array with PWM and capture/compare functions, four eight-bit input/output ports. It also has an on-chip oscillator and clock circuitry, which is operated with up to 40MHz crystal. It supports 12 clocks per machine cycle (default) and six clocks per machine cycle modes, which can be selected via the software. The Flash program memory supports both parallel programming

Test Point Detail

Test point	Details
TP0	Ground (GND)
TP1	5V
TP2	Reset when high

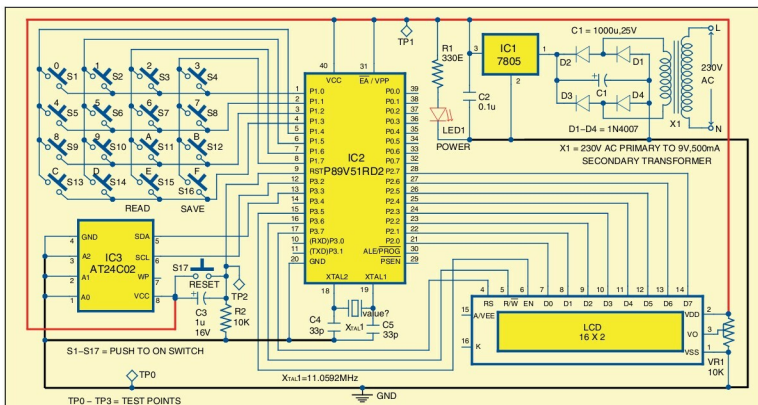


Fig. 1: Circuit of EEPROM interface for beginners